

SAP Joule

Consultant Helping Guide

Everything a consultant needs to know about SAP's Generative AI Copilot

What is Joule?

Joule Across SAP Apps

Prompting Best Practices

For Functional Consultants

For Technical Consultants

Security & Governance

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PART 1

Introduction to SAP Joule

1. What is SAP Joule?

1.1 The One-Line Answer

SAP Joule is SAP's generative AI copilot — a conversational AI assistant embedded across SAP's entire product portfolio that understands natural language, knows your SAP data and processes, and can take actions on your behalf.

Think of Joule as a highly knowledgeable SAP expert who is always available, never forgets anything, understands your role and context, and can both answer questions AND perform tasks inside SAP — all through simple, natural conversation.

1.2 Why SAP Built Joule

SAP launched Joule in October 2023 as the centerpiece of its AI strategy. The motivation was clear: SAP systems contain vast amounts of business-critical data, but accessing insights, navigating processes, and completing tasks has traditionally required deep training and expertise. Joule democratizes access to SAP by letting any user interact in plain language.

Without Joule	With Joule
Navigate 5 transactions to find a vendor's payment status	Ask: 'What is the payment status for vendor Bosch?'
Know the exact T-code or app to open for a task	Describe what you want to do in plain language
Run a report, export to Excel, analyze manually	Ask: 'Summarize this quarter's open purchase orders by category'
Submit an HR leave request through a multi-step form	Say: 'Request 3 days of annual leave starting next Monday'
Write ABAP code from scratch using documentation	Say: 'Write a CDS view for sales order items with net amount'

1.3 Joule's Core Capabilities

Answer & Explain

Joule can answer business and system questions using real-time data from your SAP applications. It understands context — if you are on the Vendor Management page, asking 'what are the open invoices?' refers to the current vendor.

Navigate & Guide

Joule can navigate you to the right SAP screen, app, or process. Instead of remembering T-codes or Fiori tile names, describe where you want to go.

Create & Draft

Joule can draft purchase orders, job descriptions, email responses, meeting summaries, financial reports, and more based on your instructions and live SAP data.

Analyze & Summarize

Joule can summarize large datasets, identify anomalies, compare periods, and surface key insights from your SAP data without requiring manual analysis.

Execute & Automate

In supported applications, Joule can perform actions — approving requests, creating records, updating fields, triggering workflows — all through conversation, with human confirmation steps where required.

1.4 Where Joule is Available (2025)

SAP Application	Joule Availability	Key Use Cases
SAP S/4HANA Cloud	Generally Available	Finance, procurement, sales, supply chain queries and tasks
SAP SuccessFactors	Generally Available	HR tasks: leave, performance, onboarding, payroll queries
SAP Ariba	Generally Available	Procurement insights, supplier queries, contract summaries
SAP Fieldglass	Generally Available	Contingent workforce management assistance
SAP Customer Experience	Available (select)	Sales, service ticket handling, customer insights
SAP Analytics Cloud	Generally Available	Natural language data queries, story generation
SAP BTP / Build Apps	Available via SDK	Custom application AI embedding via Joule SDK
SAP Concur	Roadmap	Expense and travel management assistance

SAP Datasphere	Available	Data discovery, model explanation, metadata queries
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2. How Joule Works — Under the Hood

2.1 The Technology Stack

Joule is built on SAP's AI Foundation, which runs on SAP BTP. It combines multiple AI technologies to deliver contextual, accurate, and enterprise-safe responses:

Layer	Technology	Role
Large Language Models (LLMs)	Multiple LLMs including partnerships with Mistral, Anthropic, and others	Natural language understanding, generation, reasoning
SAP AI Core	SAP's ML execution platform on BTP	Hosts and manages the AI models underlying Joule
Business Context Engine	SAP's proprietary grounding layer	Grounds LLM responses in real SAP data and metadata to prevent hallucinations
Retrieval Augmented Generation (RAG)	Vector search on SAP content + live data APIs	Fetches relevant SAP documentation, data, and configuration at query time
SAP Master Data & Transactional Data	Live connection to your SAP system's data	Ensures Joule answers are based on YOUR company's actual data
Role & Authorization Layer	XSUAA / SAP IAM integration	Ensures Joule only shows data the user is authorized to see

2.2 How a Joule Query is Processed

1. User types a natural language prompt in the Joule panel (available on the right-hand side of any SAP app)
2. Joule's intent recognition engine identifies what the user wants: a question, a navigation request, a task execution, or a content generation request
3. The context engine determines which SAP application, data objects, and user role are relevant
4. RAG retrieves relevant SAP Help content, process documentation, and live business data
5. The LLM generates a response grounded in the retrieved context and SAP data
6. The authorization layer filters the response to only include data the user is permitted to see
7. Joule presents the response with citations, data cards, and action buttons for next steps

Key Insight

Grounding

Unlike general-purpose AI (ChatGPT, Copilot), Joule's responses are always grounded in YOUR SAP data. This means Joule will not make up vendor names, order numbers, or financial figures — it reads real data from your SAP system in real time. This grounding is what makes Joule enterprise-safe.

2.3 Joule's Multi-Modal Nature

Joule is not just a text chatbot. It can interact through multiple modalities:

- **Text Chat:** The primary interface — type questions and receive answers
- **Voice Input:** Speak to Joule on mobile SAP applications (available in select apps)
- **Document Analysis:** Joule can analyze uploaded documents (contracts, invoices, reports) and answer questions about them
- **Embedded in Workflows:** Joule suggestions appear proactively at key decision points in SAP processes (e.g., approving a purchase order)
- **API-Accessible:** Developers can call Joule capabilities programmatically via BTP APIs for custom application integration

3. Joule vs. Traditional SAP Search & Navigation

3.1 The Paradigm Shift

For over 30 years, interacting with SAP required users to learn transaction codes, know menu paths, understand data structures, and navigate complex screens. Joule represents a fundamental shift from command-driven to conversation-driven interaction.

Dimension	Traditional SAP	SAP Joule
User Knowledge Required	High — must know T-codes, field names, app navigation	Low — describe intent in plain language
Data Discovery	Know where to look, run reports, export & analyze	Ask a question, get instant data-aware answer
Cross-Application Insights	Manual — open multiple apps, copy/paste data	Single question can span multiple SAP modules
Training Required	Weeks of role-specific SAP training	Hours — mostly learning effective prompting
Error Rate	High for infrequent users (wrong T-code, wrong field)	Low — Joule interprets intent and confirms before acting
Personalization	None — same UI for all users	Joule adapts to your role, recent activity, and context
Availability	Business hours + VPN + SAP GUI or browser	Always-on, mobile-friendly, same interface everywhere

For Consultants

Key Message for Client Training

When training end users on Joule, emphasize that they no longer need to memorize T-codes or navigation paths. They should describe WHAT they want to achieve, not HOW to do it in SAP. This mental shift is the biggest adoption hurdle and the biggest productivity gain.

PART 2

Joule Across SAP Applications

4. Joule in SAP S/4HANA Cloud

4.1 Accessing Joule in S/4HANA

In SAP S/4HANA Cloud, Joule is accessible via the AI Assistant panel — the small Joule icon in the top-right corner of the SAP Fiori Launchpad. Clicking it opens a slide-in panel on the right side of the screen. Joule is context-aware: it knows which app you have open and uses that context to improve response relevance.

4.2 Joule Capabilities in S/4HANA by Process Area

Finance (FI/CO)

- Query open items, account balances, and payment runs using natural language
- Summarize cost center or profit center performance for a period
- Generate draft financial journal entry descriptions
- Identify anomalies in payment behavior or budget consumption
- Navigate to the correct app for year-end closing tasks with guided steps

Procurement (MM / SRM)

- Look up purchase order status, delivery confirmations, and invoice processing status
- Draft purchase requisition descriptions and justifications
- Identify vendors with open invoices or payment blocks
- Summarize contract terms and expiry dates across the supplier portfolio
- Get guided walkthroughs for creating emergency purchase orders

Sales (SD)

- Query order status, delivery schedules, and billing documents for any customer
- Identify customers with overdue payments or blocked orders
- Draft customer-facing order confirmation or delay notification emails
- Summarize sales performance by region, product line, or sales rep
- Get guided steps for processing returns and credit memos

Supply Chain (PP/WM/TM)

- Query production order status and material availability
- Identify bottlenecks in production scheduling or warehouse operations
- Summarize open delivery backlogs and shipment delays
- Generate exception reports for materials at risk of stockout

4.3 Sample S/4HANA Prompts

Finance — AR Aging Query

You: Show me the top 10 customers with the highest overdue receivables this month

Joule: Here are the top 10 customers by overdue receivables as of today. Customer XYZ Corp leads with EUR 245,000 overdue (45+ days). Would you like me to generate a collections priority report or draft follow-up emails for these customers?

Procurement — PO Status Check

You: What purchase orders for vendor Siemens are pending goods receipt?

Joule: I found 7 open purchase orders for Siemens with pending GR. The oldest is PO 4500012345 from March 2nd for EUR 82,000. 3 of these are past their expected delivery date. Shall I show you the details or create a follow-up message to the vendor?

Sales — Blocked Order Resolution

You: Why is sales order 1234567 blocked and what do I need to do to release it?

Joule: Sales order 1234567 is blocked due to a credit limit exceeded status. Customer ABC Ltd has EUR 150,000 outstanding against a EUR 200,000 credit limit, and this order for EUR 75,000 would exceed it. To release: either get a credit limit increase approved in FD32, or request a payment from the customer first. Would you like me to initiate the credit limit review workflow?

5. Joule in SAP SuccessFactors

5.1 Overview

Joule is deeply embedded in SAP SuccessFactors for HR scenarios. It serves both HR professionals managing the workforce and employees managing their own employment-related tasks. SuccessFactors is one of the most mature Joule integrations.

5.2 Joule Capabilities in SuccessFactors

Module	Joule Capability
Employee Central	Query employee data, org charts, headcount; update personal information; navigate HR processes
Time & Attendance	Request leave, check leave balance, view team calendar, submit timesheets via conversation
Performance & Goals	Draft performance review comments, summarize team performance, set goal templates
Learning	Find relevant learning courses, summarize course content, track completion status
Recruiting	Draft job descriptions, summarize candidate profiles, compare applicants
Onboarding	Guide new hires through onboarding steps, answer policy questions, complete paperwork
Compensation	Query salary bands, summarize pay equity analysis, model compensation scenarios
Succession	Identify succession candidates, summarize talent profiles, generate development plans

5.3 Sample SuccessFactors Prompts

HR Manager — Team Leave Overview

You: Who on my team is on leave next week?

Joule: Next week, 3 members of your team are on approved leave: Sarah Johnson (Mon-Wed, Annual Leave), Marcus Lee (Tue-Fri, Medical Leave), and Priya Sharma (Mon, Personal Day). Your team will be at 75% capacity. Would you like to see the full team calendar or arrange coverage?

Recruiter — Job Description Drafting

You: Draft a job description for a Senior SAP BTP Developer with 5+ years experience in CAP and Kyma

Joule: *Here is a draft job description for Senior SAP BTP Developer: [full JD generated including role overview, key responsibilities, required qualifications, preferred skills, and company benefits placeholder]. This is based on the role template in your system and current market benchmarks. Shall I refine any section or post it directly to your open requisition?*

6. Joule in SAP Ariba

6.1 Overview

In SAP Ariba, Joule assists procurement professionals and buyers with sourcing, contract management, and supplier collaboration. Joule can analyze supplier risk, summarize contract terms, and guide buyers through complex procurement processes.

6.2 Key Capabilities in Ariba

- Supplier Intelligence: 'What is the financial risk rating for our top 10 suppliers in Asia Pacific?'
- Contract Summary: 'Summarize the key terms and renewal dates for our top 5 IT contracts'
- Spend Analysis: 'Show me maverick spend in the IT category for this quarter'
- Sourcing Guidance: 'What is the recommended sourcing strategy for industrial lubricants?'
- RFP Drafting: 'Draft an RFP template for cloud infrastructure services'
- Compliance Check: 'Which suppliers are past their sustainability certification renewal date?'

7. Joule in SAP BTP & SAP Build

7.1 Joule as a Development Accelerator

For technical users on SAP BTP, Joule is embedded in SAP Business Application Studio (BAS) and SAP Build as an intelligent coding and development assistant. It understands SAP-specific frameworks (CAP, RAP, Fiori) and can generate production-ready code.

7.2 Joule in Business Application Studio

- **Code Generation:** Generate CDS entity definitions, OData service handlers, ABAP RAP classes from natural language descriptions
- **Code Explanation:** Highlight any code block and ask Joule to explain what it does in plain English
- **Code Review:** Ask Joule to review your code for errors, performance issues, or security vulnerabilities
- **Documentation:** Generate inline code comments and API documentation from existing code
- **Unit Test Generation:** Ask Joule to write unit tests for a given service handler or method

Developer — CAP CDS Generation

You: Generate a CAP CDS entity for a Travel Request with fields for employee, destination, travel dates, estimated cost, currency, and approval status

Joule: Here is the CDS entity definition for TravelRequest: `[entity TravelRequest { key ID: UUID; employee: Association to Employees; destination: String(100); departureDate: Date; returnDate: Date; estimatedCost: Decimal(15,2); currency: String(5); status: String(20) enum { draft; submitted; approved; rejected; }; } ]` Shall I also generate the service definition, sample data, and Fiori annotations for a List Report?

7.3 Joule in SAP Build Process Automation

- **Process Modelling Assistance:** Describe a business process in plain language and Joule suggests the workflow structure
- **Decision Table Generation:** Ask Joule to generate decision logic tables based on business rules
- **Bot Script Generation:** Describe a UI automation task and Joule generates the RPA bot script steps

8. Joule in SAP Analytics Cloud

8.1 Natural Language Analytics

In SAP Analytics Cloud, Joule enables users to query data in plain English without building reports. This makes analytics accessible to every business user, not just trained BI analysts.

8.2 Key Capabilities in SAC

- Smart Discovery: 'What are the key drivers of revenue decline in the EMEA region this quarter?'
- Story Generation: 'Create a dashboard story showing sales performance vs target by region'
- Anomaly Explanation: 'Why is there a spike in operating expenses in Q3 compared to the forecast?'
- Forecast Queries: 'What is the predicted cash flow for the next 6 months based on current trends?'
- Natural Language Charts: 'Show me a bar chart of gross margin by product category for 2024'

9. Joule in SAP Customer Experience (CX)

9.1 Sales & Service Scenarios

In SAP Sales Cloud and SAP Service Cloud, Joule assists sales reps and service agents with customer-facing tasks and internal knowledge retrieval.

CX Module	Joule Use Case
SAP Sales Cloud	Summarize account history before a meeting; draft follow-up emails; identify upsell opportunities based on purchase history
SAP Service Cloud	Suggest resolution steps for support tickets; summarize ticket history for a customer; draft response emails to customers
SAP Commerce Cloud	Analyze conversion funnel drop-off; recommend product catalog optimizations; identify pricing anomalies
SAP Marketing Cloud	Summarize campaign performance; draft marketing copy variations; analyze audience segment behavior

PART 3

Effective Prompting for Consultants

10. The Art of Prompting Joule

10.1 Why Prompting Matters

The quality of Joule's response depends heavily on the quality of the prompt. A vague prompt produces a vague answer. A well-crafted prompt that includes context, role, data scope, and desired output format produces a precise, actionable response. As a consultant, mastering prompting both for yourself and for training end users is a critical skill.

10.2 The CRAFT Prompting Framework

For SAP Joule, we recommend the CRAFT framework for constructing effective prompts:

Letter	Element	What to Include	Example
C	Context	What is your current situation or starting point?	'I am reviewing the monthly close for October...'
R	Role	What is your role or perspective?	'As the AP Manager...' or 'As the project lead...'
A	Action	What do you want Joule to do?	'...summarize...', '...draft...', '...find...', '...create...'
F	Format	How should the response be formatted?	'...in a table', '...as a bullet list', '...in an email format'
T	Target	What is the scope, filter, or time period?	'...for vendor SAP SE', '...for this quarter', '...top 10 only'

CRAFT in Action — Example

CRAFT Framework Applied

You: As the Procurement Manager (R), I am preparing for the monthly supplier review (C). Please summarize (A) in a table format (F) the top 10 vendors by purchase order volume for Q3 2024, showing PO count, total value, and on-time delivery rate (T).

Joule: Here is the Q3 2024 top vendor summary table: [Table showing Vendor Name | PO Count | Total Value (EUR) | On-Time Delivery %]. Bosch leads with 145 POs worth EUR 2.3M at 94% on-time. 3 vendors have below 80% on-time delivery — would you like me to flag these for your review agenda?

10.3 Joule Prompt Types

Prompt Type	Purpose	Pattern	Example
Lookup/Query	Retrieve specific data or status	'What is the [data] for [object/entity]?'	'What is the credit limit for customer BMW?'
Summary	Condense large data into key insights	'Summarize [topic/data] for [scope/period]'	'Summarize open purchase orders for this week'
Comparison	Compare two periods, entities, or scenarios	'Compare [A] vs [B] for [metric]'	'Compare sales in Q3 2024 vs Q3 2023 by region'
Navigation	Find the right screen or app	'How do I [task] in SAP?'	'How do I create an intercompany transfer in S/4HANA?'
Draft/Generate	Create documents, emails, descriptions	'Draft a [document type] for [context]'	'Draft an email to vendor Siemens about a delayed delivery'
Analysis	Identify patterns, anomalies, root causes	'Why is [metric] showing [anomaly]?'	'Why did our gross margin drop 3% in October?'
Action	Execute a task in SAP	'Please [action] for [object]'	'Please create a leave request for 3 days starting Monday'
Explain	Understand a process, field, or error	'Explain [concept/error/field]'	'Explain what the payment block reason B means in FI'

11. Prompt Templates by Business Process

11.1 Finance Prompt Templates

Scenario	Recommended Prompt Template
Month-end close status	'Summarize the status of all open month-end closing tasks for [company code] as of today. Flag any overdue items.'
Budget vs. Actual	'Compare actual spend vs budget for cost center [CC number] for [period]. Highlight areas exceeding 90% of budget.'
Cash flow analysis	'What is the current cash position and projected cash flow for the next 30 days for company code [CC]?'
Overdue receivables	'List all customer invoices overdue by more than [X] days for [company code]. Show amount, customer name, and days overdue.'
Payment run preparation	'Which open vendor invoices are due for payment this week? Group by payment method and summarize total by currency.'
GR/IR reconciliation	'Identify all GR/IR clearing account items older than 30 days that have not been resolved. Show PO number, vendor, and amount.'

11.2 Procurement Prompt Templates

Scenario	Recommended Prompt Template
PO status overview	'Show me all purchase orders for [vendor/category] that are open and past their expected delivery date.'
Spend analysis	'What was the total spend in the [category] category for [period]? Compare to the previous year and highlight the top 5 vendors.'
Contract expiry alerts	'Which supplier contracts are expiring in the next 90 days? Include contract value and responsible buyer.'
Maverick spend	'Identify any purchases made in the last quarter without an approved purchase order. Show department, amount, and vendor.'
Supplier performance	'Summarize the delivery performance of our top 10 vendors for this year. Show on-time delivery rate and quality rejection rate.'
Emergency PO guidance	'What is the process for creating an emergency purchase order for a critical production material in S/4HANA?'

11.3 HR / SuccessFactors Prompt Templates

Scenario	Recommended Prompt Template
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Headcount summary	'Provide a headcount summary for [department/region] as of today. Include full-time, part-time, and open positions.'
Leave management	'Which employees in [team] have not taken any leave in the past 6 months? Show leave balance remaining.'
Performance cycle	'Summarize the performance review completion status for [department]. Who has not submitted their self-assessment?'
Onboarding status	'What onboarding tasks are still incomplete for new hires who joined in [month]? Show employee name and pending tasks.'
Job description drafting	'Draft a job description for a [role title] in the [department] team requiring [X years] experience with [skills].'
Policy question	'What is our company policy on parental leave duration and eligibility criteria?'

11.4 Analytics / SAC Prompt Templates

Scenario	Recommended Prompt Template
Revenue analysis	'What were the top 5 revenue-generating products in [region] for [quarter]? Compare to the same period last year.'
Anomaly detection	'Are there any unusual patterns in our sales data for the past 30 days? Flag any products with more than 20% deviation from forecast.'
Forecast review	'Show me the revenue forecast vs actuals for the full year. Which business units are most at risk of missing their targets?'
Executive summary	'Generate a one-page executive summary of this month's financial performance for the board presentation.'

12. Prompt Dos and Don'ts

DO	DON'T
Be specific about the data scope: 'for Q3 2024, company code 1000'	Be vague: 'show me sales data'
Specify the output format: 'in a table', 'as a bullet list', 'as an email'	Leave format unspecified when you want a specific structure
Provide business context: 'I am preparing for a supplier review meeting'	Assume Joule knows what you are doing without telling it
Ask follow-up questions to refine responses: 'Now sort by highest risk first'	Abandon a prompt and start over when the first answer is close but not perfect
Use role context: 'As the AP accountant...' or 'As the hiring manager...'	Ignore your role context — it helps Joule understand what data is relevant
Break complex requests into steps: first ask for data, then ask for analysis	Ask for 5 things in one prompt — Joule handles focused requests better
Ask Joule to explain its reasoning: 'Why did you identify this as an anomaly?'	Accept answers without understanding the basis for them
Save effective prompts as templates for your team to reuse	Re-type the same good prompts from scratch every time

Consultant Tip

Build a Prompt Library

During any Joule implementation project, maintain a living 'Prompt Library' document — a collection of validated, effective prompts organized by role and process area. This becomes a key deliverable of the project and dramatically accelerates end-user adoption. Share it as part of the training package.

PART 4

Functional Consultant's Joule Guide

13. Joule for SD / Order-to-Cash Consultants

13.1 How Joule Transforms O2C Processes

The Order-to-Cash (O2C) process is one of the most complex and data-intensive in SAP. Sales consultants working on S/4HANA implementations should understand how Joule changes the way end users interact with sales, shipping, and billing processes.

O2C Step	Traditional SAP Interaction	With Joule
Inquiry / Quote	Create manually in VA11/VA21 after looking up pricing conditions	'Create a quotation for customer [name] for [product] — pull current pricing'
Sales Order	VA01 with multiple tab navigation and field entry	'Create a sales order for customer BMW for 100 units of product X001'
Credit Check	FD32 to check credit, FD33 to view — separate transactions	'Why is customer XYZ's order blocked? What is their credit utilization?'
Delivery	VL01N — check availability, create delivery manually	'What deliveries are due for shipment today? Are any at risk of delay?'
Billing	VF01 — billing due list, individual or collective billing	'Which deliveries are ready for billing and not yet invoiced?'
Collections	FBL5N — open item report + manual follow-up	'Draft collection emails for the 5 most overdue customer accounts'

13.2 SD-Specific Joule Use Cases for Consultants

During Implementation

- Use Joule to generate test scripts: 'Generate a test script for the end-to-end OTC process with credit check'
- Ask Joule to explain SAP standard pricing procedure logic: 'Explain how condition type PR00 works in S/4HANA pricing determination'
- Use Joule in BAS to generate CAP extensions for custom SD business logic

During Training

- Use Joule to demonstrate queries without needing test data — Joule can explain what a response would look like
- Create process walkthroughs using Joule's navigation guidance feature
- Use Joule's explain mode to help users understand why certain blocks or errors occur

During Go-Live Support

- Use Joule to quickly diagnose production issues: 'Why is output not being triggered for sales orders created today?'
- Ask Joule to identify systemic issues: 'Are there patterns in the sales orders that failed credit check today?'

14. Joule for MM / Procure-to-Pay Consultants

14.1 P2P Process with Joule

P2P Step	Joule Capability
Purchase Requisition (PR)	Create PRs via natural language; check if similar PRs exist; get approval status
RFQ / Quotation	Draft RFQ content; compare vendor quotations; summarize price differences
Purchase Order (PO)	Create POs for approved PRs; check PO status; identify POs with missing confirmations
Goods Receipt (GR)	Check expected GR dates; identify overdue deliveries; get guided steps for GR posting
Invoice Verification (IV)	Identify invoices pending 3-way match; explain payment blocks; find GR/IR variances
Payment	Check payment run schedule; find invoices due for next payment run; explain payment terms
Supplier Management	Summarize supplier performance; flag high-risk suppliers; draft supplier communications

14.2 MM Consultant Value-Add with Joule

- Use Joule to automatically draft Material Master descriptions from technical specifications
- Generate configuration documentation by asking Joule to explain the current purchasing org setup
- Accelerate UAT by using Joule to generate test case descriptions for P2P scenarios
- Create supplier communication templates using Joule's drafting capability during implementation

15. Joule for FI/CO / Finance Consultants

15.1 Finance-Specific Joule Capabilities

Finance Area	Joule Use Case	Sample Prompt
General Ledger	Query account balances, explain postings, identify unusual transactions	'Show all GL postings to account 400000 in October that are above EUR 100K'
Accounts Payable	Check invoice status, explain payment blocks, prepare payment run	'Which invoices from vendor Siemens have a payment block and why?'
Accounts Receivable	Aging analysis, collection prioritization, dunning status	'Show AR aging analysis for company code 1000 as of today'
Asset Accounting	Query asset values, depreciation runs, asset retirement	'What assets are due for depreciation run this month? Show net book value.'
Controlling	Cost center variance analysis, internal order tracking, profitability	'Compare plan vs actual for all cost centers in the IT department for this year'
Treasury	Cash position, bank account balances, FX exposure	'What is our net FX exposure in USD as of today across all company codes?'
Tax	Explain VAT postings, identify missing tax codes, support tax returns	'Which vendor invoices are missing a tax code in the current month?'

15.2 Month-End Close Acceleration with Joule

Month-end close is one of the most time-pressured processes in finance. Joule can significantly reduce close time by providing instant status visibility and automated drafting capabilities.

- Close Status Dashboard: 'Give me a complete status of all open month-end closing activities for all company codes'
- Intercompany Matching: 'Identify all intercompany transactions that are not yet matched and reconciled'
- Journal Entry Support: 'Draft the journal entry description for the Q3 accrual for IT services'
- Variance Commentary: 'Generate variance commentary for the P&L comparing October actuals vs budget'

16. Joule for HCM / HR Consultants

16.1 HR Process Transformation with Joule

Human Capital Management is one of the areas where Joule has the highest employee-facing impact. HR consultants implementing SuccessFactors should position Joule as the single access point for all employee HR needs.

HR Process	Joule Capability	User Impact
Leave Management	Apply for leave, check balance, view team calendar via conversation	Eliminates 3-4 screen navigation for self-service
Payslip / Compensation	Query payslip details, understand deductions, check benefits enrollment	Reduces HR helpdesk tickets by up to 40%
Performance Reviews	Get reminders, draft review narratives, understand rating scales	Faster review completion, more consistent narratives
Policy Questions	Ask HR policy questions in plain language (parental leave, travel policy)	Self-service policy compliance without emailing HR
Learning & Development	Find courses, enroll, summarize learning content	Higher L&D engagement and course completion rates
Onboarding	Guide new employees through tasks, answer questions, connect to buddy	Better first-day experience, faster time-to-productivity
Org Chart Navigation	Find colleagues, understand reporting lines, get contact info	Faster internal navigation for new employees

HR Consultant Insight	Joule as the HR Service Desk Position Joule to clients as a first-line HR service desk that handles the 60-70% of routine employee queries (leave balance, payslip, policy questions). This frees HR Business Partners to focus on strategic work. Frame the ROI around HR helpdesk ticket reduction and employee satisfaction scores.
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PART 5

Technical Consultant's Joule Guide

17. Joule for ABAP & BTP Developers

17.1 Joule as a Development Assistant

For ABAP and BTP developers, Joule in Business Application Studio is a force multiplier. It understands SAP-specific frameworks at a deep level — it knows CDS syntax, RAP patterns, CAP model conventions, and Fiori annotations. This makes it fundamentally more useful for SAP developers than general coding assistants.

17.2 Joule Coding Capabilities in BAS

Task	What Joule Can Do	Time Saved
CDS View Generation	Generate complete CDS view with associations, annotations, and access controls from a description	2-4 hours → 15 minutes
RAP BDEF Generation	Generate Behavior Definition for managed or unmanaged BOs with operations, validations, and actions	3-5 hours → 30 minutes
CAP Handler Logic	Generate Node.js or Java service handler with CRUD logic, validations, and error handling	2-3 hours → 20 minutes
iFlow Design Advice	Explain the best CPI adapter and message mapping approach for a given integration scenario	Research time eliminated
UI5 Component Generation	Generate SAPUI5 XML views and controllers for standard list, detail, and form patterns	4-6 hours → 45 minutes
Unit Test Generation	Generate comprehensive unit tests for CAP handlers, covering happy path and error scenarios	2-3 hours → 20 minutes
Code Review	Analyze code for performance issues, security vulnerabilities, and SAP best practice violations	Peer review time saved

Documentation	Generate technical documentation, API reference, and inline comments from existing code	1-2 hours → 5 minutes
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17.3 Advanced Developer Prompts

RAP Development

You: Generate a managed Business Object for Equipment Maintenance with header (Equipment, Description, PlantCode, Category) and item entities (MaintenanceItem, TaskDescription, PlannedDate, Status). Include validations for mandatory fields and a custom action to confirm maintenance completion.

Joule: Here is the complete RAP Business Object definition including: Root Entity CDS view with @OData.publish, Child Entity CDS view with composition, BDEF with managed implementation, validations for mandatory fields on SAVE, and a confirmMaintenance action. I have also included the Projection BDEF and suggested Service Definition. Shall I generate the Behavior Implementation class stubs and the Service Binding?

Performance Troubleshooting

You: My CAP service query on SalesOrderItems is taking 8 seconds. Here is the CDS view definition: [paste CDS]. What are the performance issues and how should I fix them?

Joule: I can see 3 potential performance issues in your CDS view: 1) Missing @AbapCatalog.buffering on the association to Material — add buffering for master data associations. 2) The WHERE clause on VBAP.ERDAT lacks an index — consider adding @AbapCatalog.compiler.compareFilter. 3) The nested subquery for pricing conditions should be flattened into a JOIN. Here is the optimized CDS view: [optimized code]. Expected improvement: 60-75% query time reduction.

18. Embedding Joule in Custom Applications

18.1 Joule SDK Overview

SAP provides the Joule SDK for BTP, allowing developers to embed Joule's AI capabilities directly into custom CAP applications, SAPUI5 apps, and third-party applications. This enables organizations to build custom workflows with Joule-powered intelligence.

18.2 Integration Approaches

Approach	Description	Best For
Joule Floorplan (UI5)	Pre-built SAPUI5 component that embeds the full Joule chat panel in custom Fiori apps	Adding Joule to existing Fiori apps quickly
Joule API (BTP)	REST API that allows any application to send prompts to Joule and receive structured responses	Backend processing, non-Fiori apps, mobile apps
Joule Skills SDK	Framework for building custom Joule 'skills' that extend Joule's capabilities to custom data sources	Deep custom application integration
SAP AI Core API	Direct access to the underlying LLM models for fully custom AI implementations	Novel AI features beyond standard Joule scope

18.3 Custom Joule Skill Architecture

When embedding Joule in custom applications, the technical architecture involves:

8. User sends a natural language request via the Joule panel embedded in your custom app
9. Joule's intent recognition identifies the request relates to your custom skill
10. The skill definition routes the request to your custom CAP or microservice backend
11. Your backend executes business logic, queries custom or SAP data, and returns structured results
12. Joule formats the results in natural language and presents them to the user
13. Optional: Joule presents action buttons for follow-up tasks

19. Joule Studio — Building Custom AI Skills

19.1 What is Joule Studio?

Joule Studio is SAP's low-code/no-code environment on BTP for building custom AI skills that extend Joule's capabilities to your organization's specific processes, data, and terminology. It allows business analysts and developers to create custom intents, actions, and responses without deep AI expertise.

19.2 Key Joule Studio Concepts

Concept	Definition
Skill	A custom capability added to Joule. A skill defines what Joule can understand and do in a specific domain (e.g., 'Fleet Management skill' or 'Custom HR Policy skill')
Intent	A specific goal or task a user might request within a skill (e.g., 'Check vehicle service due date', 'Request fleet vehicle booking')
Entity	A data element extracted from user input (e.g., vehicle registration number, date, employee name)
Action	What Joule does when an intent is recognized — calls an API, queries a database, triggers a workflow
Response Template	How Joule formats and presents the result of an action to the user
Knowledge Base	Custom documents, policies, FAQs, and content that Joule uses to answer questions in your domain

19.3 Step-by-Step: Building a Custom Skill

14. Define the skill scope: What domain does this skill cover? What questions should it answer? What actions should it enable?
15. Create the skill in Joule Studio: Name it, describe it, and assign it to the relevant SAP application context
16. Define intents: List the top 10-20 user requests this skill should handle. Provide example utterances for each intent
17. Configure entities: Define the key data points to extract from user prompts (dates, names, IDs, categories)
18. Connect actions: Link each intent to a backend API call, OData query, or BTP workflow trigger
19. Add knowledge base content: Upload FAQs, policy documents, process guides for RAG-based question answering
20. Test in Joule Studio: Use the built-in simulator to test intents, entity extraction, and action execution

21. Deploy to subaccount: Publish the skill to your BTP subaccount and assign to user groups via Role Collections

PART 6

Security, Governance & Adoption

20. Data Privacy & Security in Joule

20.1 How SAP Protects Data in Joule

Data security is one of the first concerns raised by clients when introducing Joule. Consultants must be equipped to address these concerns clearly and accurately. SAP has built enterprise-grade security into Joule at every layer.

Security Concern	How SAP Addresses It
Is my data sent to OpenAI or public LLMs?	No. Joule uses SAP AI Core which runs LLMs in SAP's own tenanted infrastructure. Your data does not leave SAP's data centers or go to public AI providers unless explicitly configured.
Can users see data they are not authorized to see?	No. Joule respects all existing SAP authorization checks. It only retrieves and displays data that the logged-in user's role and authorizations permit.
Is conversation history stored? Who can access it?	Conversation history is stored per user in an encrypted, tenant-isolated store. SAP support does not have access to user conversations without explicit consent.
Does Joule learn from our data and share it with other tenants?	No. Joule does not use your company's data to train shared models. Each tenant's data is strictly isolated.
What about GDPR compliance for personal data in prompts?	SAP's AI services comply with GDPR. Data residency options are available by selecting the appropriate BTP data center region. See SAP's Data Processing Agreement.
Can Joule make unauthorized changes to our SAP system?	No. Joule can only execute actions that are within the user's authorization scope. Destructive or major actions require explicit user confirmation before execution.

20.2 Governance Framework for Joule

Organizations implementing Joule should establish a clear governance framework:

- **AI Usage Policy:** Define acceptable use cases for Joule and communicate what employees should and should not use it for
- **Data Classification:** Define which data categories can be referenced in Joule prompts (e.g., no customer PII in prompt text)

- **Audit Trail:** Enable SAP Audit Log Service to capture Joule usage events for compliance review
- **Joule Access Control:** Control which users have access to Joule via Role Collections in BTP Cockpit
- **Custom Skill Review Process:** Establish a review and approval process before custom skills are deployed to production
- **AI Response Validation:** Train users to validate Joule's outputs for critical decisions — do not blindly execute AI suggestions

21. Change Management & User Adoption

21.1 The Adoption Challenge

Technology adoption is often the hardest part of a Joule rollout. Many users are skeptical of AI, some are intimidated by conversational interfaces, and others are concerned about job security. A structured change management approach is essential.

21.2 The Joule Adoption Journey

Phase	Duration	Focus	Key Activities
Awareness	Weeks 1-2	Create excitement and understanding	Executive sponsorship messages, Joule demo videos, town halls, FAQ sessions
Exploration	Weeks 3-4	Let users experiment safely	Sandbox access, lunch & learn sessions, self-paced Joule tutorials, pilot group selection
Learning	Weeks 5-8	Build prompting skills	Role-specific training sessions, prompt library creation, 'Joule Champion' program
Adoption	Weeks 9-12	Embed in daily workflows	Process walk-throughs with Joule, manager reinforcement, usage tracking, feedback loops
Optimization	Ongoing	Maximize value and expand	New skill deployment, advanced prompting workshops, ROI measurement, expansion planning

21.3 Joule Champion Program

The most effective adoption strategy is identifying and empowering 'Joule Champions' — enthusiastic early adopters in each business area who become local experts and peer coaches.

- Select 1-2 champions per business unit or department
- Give champions early access and advanced training
- Empower them to build and share their own prompt libraries
- Create a champions community channel (Teams/Slack) for sharing tips and wins
- Recognize champion contributions in company communications

21.4 Training Delivery Best Practices

- Keep training sessions short and role-specific — 60-90 minutes maximum, focused on the user's actual job tasks
- Use real company data examples (anonymized where needed) rather than generic demos
- Give every attendee a printed or digital Prompt Starter Card with 10 prompts relevant to their role

- Record demo sessions and share as on-demand resources
- Create a Joule FAQ page in the company intranet addressing the most common concerns

22. Measuring Joule ROI & Success Metrics

22.1 Quantitative Metrics

Metric Category	KPI	Measurement Method
Productivity	Time saved per user per week on routine SAP tasks	Pre/post time-motion studies or user self-reporting surveys
Adoption	% of licensed users actively using Joule monthly	SAP Joule usage analytics dashboard
Service Desk	Reduction in IT/HR helpdesk tickets for 'how do I' questions	Ticketing system comparison pre/post rollout
Quality	Reduction in data entry errors (wrong T-code, wrong field)	Error rate tracking in key process KPIs
Speed	Reduction in process cycle time (e.g., purchase requisition to PO)	Process mining data from SAP Signavio or SAP Analytics
Training	Reduction in new user onboarding time to system proficiency	HR / L&D metrics on time-to-competency
Finance	Reduction in month-end close cycle time	Finance operations dashboards

22.2 Qualitative Metrics

- User satisfaction scores (1-5 rating after Joule interactions)
- Employee NPS (Net Promoter Score) for IT systems — benchmark before and after
- Manager feedback on team productivity and focus on value-added work
- Number of new use cases identified by champions and business users

23. Joule Roadmap & Future Capabilities

23.1 What is Coming (SAP's Publicly Stated Direction)

SAP has committed to making Joule the primary interface for all SAP applications by 2026. Key roadmap themes include:

- **Agentic AI:** Joule evolving from a question-answer assistant to an autonomous agent that can execute multi-step business processes end-to-end without step-by-step human guidance
- **Cross-Application Orchestration:** Joule completing tasks that span multiple SAP applications (e.g., create a PO in S/4HANA, notify the supplier via Ariba, update the budget in SAC — all from one prompt)
- **Proactive Intelligence:** Joule proactively alerting users to exceptions, risks, and opportunities without being asked — 'push' rather than 'pull' AI
- **Voice-First Interfaces:** Native voice input across all SAP mobile applications powered by Joule
- **Multi-Modal Analysis:** Joule analyzing images, videos, and documents (e.g., 'Analyze this machinery inspection photo and log a maintenance ticket')
- **Industry-Specific Skills:** Pre-built Joule skills for specific industries (retail demand planning, pharma compliance, utilities grid management)

Consultant Advisory

Position Joule as a Journey

Advise clients to adopt Joule as a platform for continuous AI adoption, not a one-time project. The capabilities available today are the foundation — the real value compounds as Joule's agentic capabilities mature. Clients who build strong foundations now (governance, prompt culture, data quality) will have a significant competitive advantage as Joule capabilities expand.

24. Quick Reference — Prompt Cheat Sheet

Use this reference card during discovery workshops, training sessions, and daily use:

Prompt Intent	Starter Phrase	Example
Check Status	'What is the status of...'	'What is the status of PO 4500098765?'
Find Information	'Show me / Find / List...'	'Show me all overdue invoices for company code 1000'
Summarize Data	'Summarize / Give me a summary of...'	'Summarize Q3 procurement spend by category'
Compare Periods	'Compare [A] vs [B]...'	'Compare headcount in Engineering this year vs last year'
Create / Draft	'Create / Draft / Write...'	'Draft a supplier communication about the delayed shipment on PO 123'
Explain	'Explain / Why is...'	'Why is this sales order blocked? Explain the reason.'
Navigate	'How do I / Where do I...'	'How do I post a manual journal entry in S/4HANA?'
Analyze	'Analyze / Identify patterns in...'	'Analyze our vendor payment behavior — are we paying early or late?'
Take Action	'Please [action]...'	'Please approve the leave request from John Smith for next week'
Generate Report	'Generate a report / Create a summary...'	'Generate a cost center variance report for the IT department for October'
Ask What-If	'What would happen if...'	'What would happen to our DSO if we reduce payment terms from Net 45 to Net 30?'
Get Recommendations	'What do you recommend / Suggest...'	'What should I prioritize in the purchase order backlog today?'

End of Document

SAP Joule — Consultant Helping Guide

For more: help.sap.com/joule | community.sap.com | [SAP Learning Hub](#)